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PROJECT OVERVIEWS

Project Name: Modular Structured-Light 3D Reconstruction Pipeline
Course: COMPSCI 117 – Project in Computer Vision
Instructor: Charles Fowlkes

Objective:
To design a modular 3D reconstruction pipeline that automatically processes structured-light Gray-code pattern images captured from multiple angles, enabling precise mesh generation. The project focused on minimizing depth noise and background artifacts while achieving robust multi-scan registration.

TECHNICAL CHALLENGES

Core Challenges:

Minimizing noise and background interference during Gray-code depth estimation and achieving stable alignment across multi-view scans.

Issues Identified:

- Noise and artifacts introduced during depth extraction
- Difficulty separating objects from background boundaries
- Instability during multi-view scan alignment

Strategies Implemented:

- Motion-, color-, and mask-based 3D filtering
- SVD-based Procrustes alignment for automated scan registration
- Poisson surface reconstruction for watertight mesh generation
- Modular staged process: Preprocessing → Alignment → Mesh Generation

CONTRIBUTION

Modular Architecture:

Separated into helpers.py, camutils.py, meshutils.py for better maintainability and debugging.

Core Algorithm Implementations:

- Gray-code based bit pattern decoding + 3-way masking (motion + color + consistency)
- Mesh denoising with Laplacian smoothing and bounding-box pruning
- Automated multi-view alignment via SVD-based Procrustes transformation
- Watertight mesh creation through Poisson surface reconstruction

Automation & Optimization:

- Single-run pipeline executable via driver.py
- Intermediate caching for reduced processing time and batch execution stability

OUTCOME

Achievements:

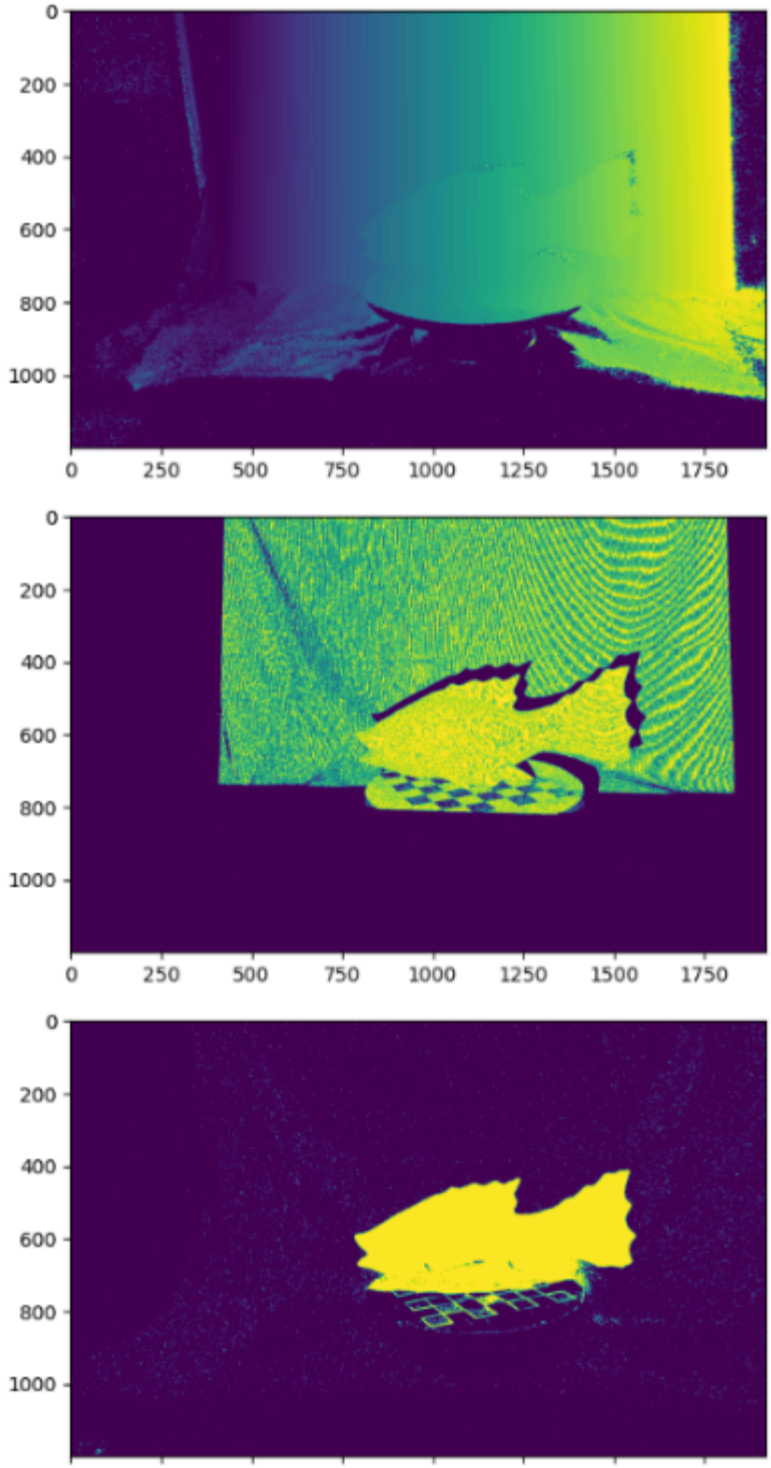
- Successfully produced fully aligned 3D meshes from 5 viewing angles
- Reduced background noise and shadows for accurate shape preservation
- Achieved smooth and uniform watertight final meshes after Poisson reconstruction

Visual Results:

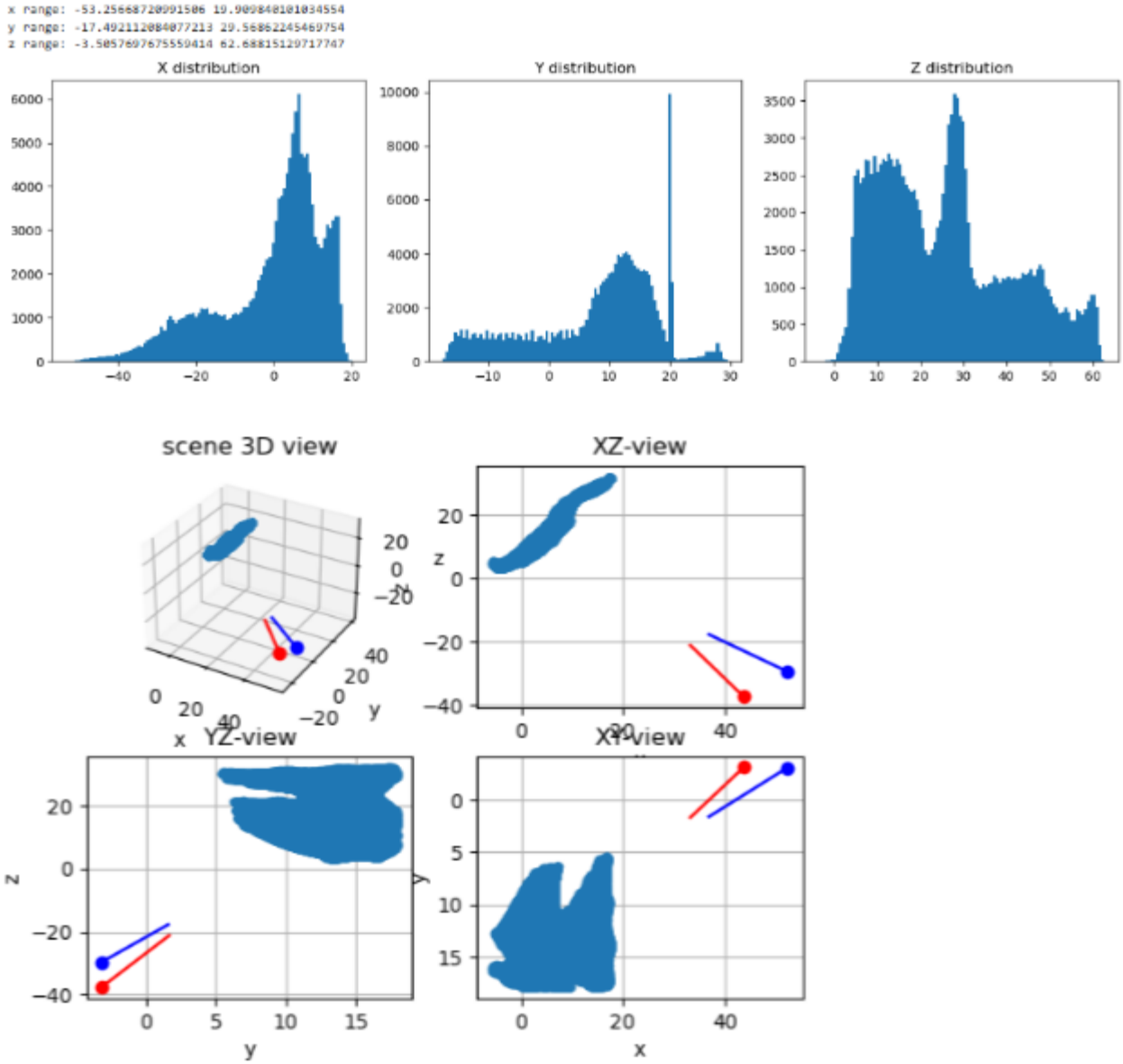
- Visualization across pipeline stages: Gray-code Decoding → Triangulation → Mesh Cleaning → Merging

- Preserved fine surface details such as fins and scales

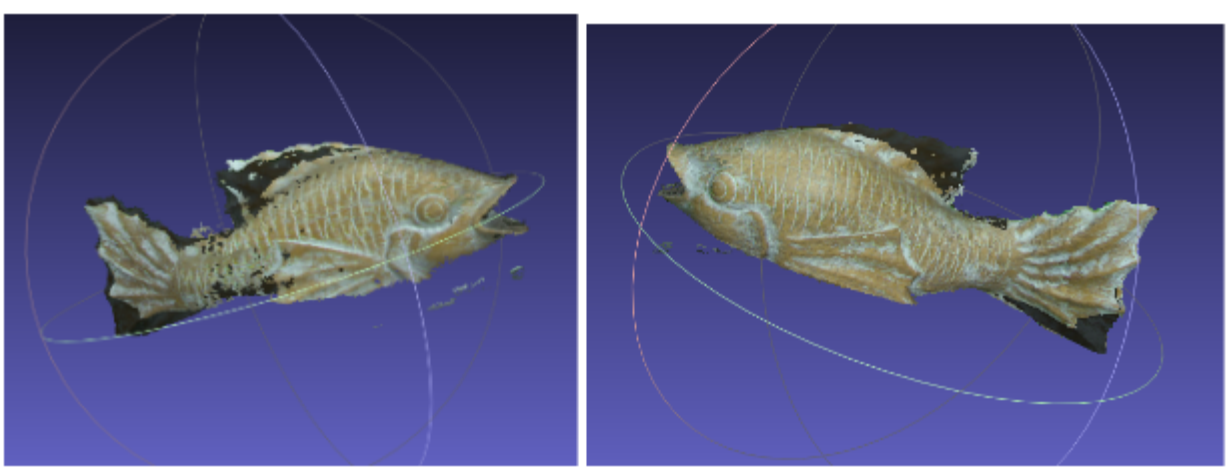
3-way Masking Results:



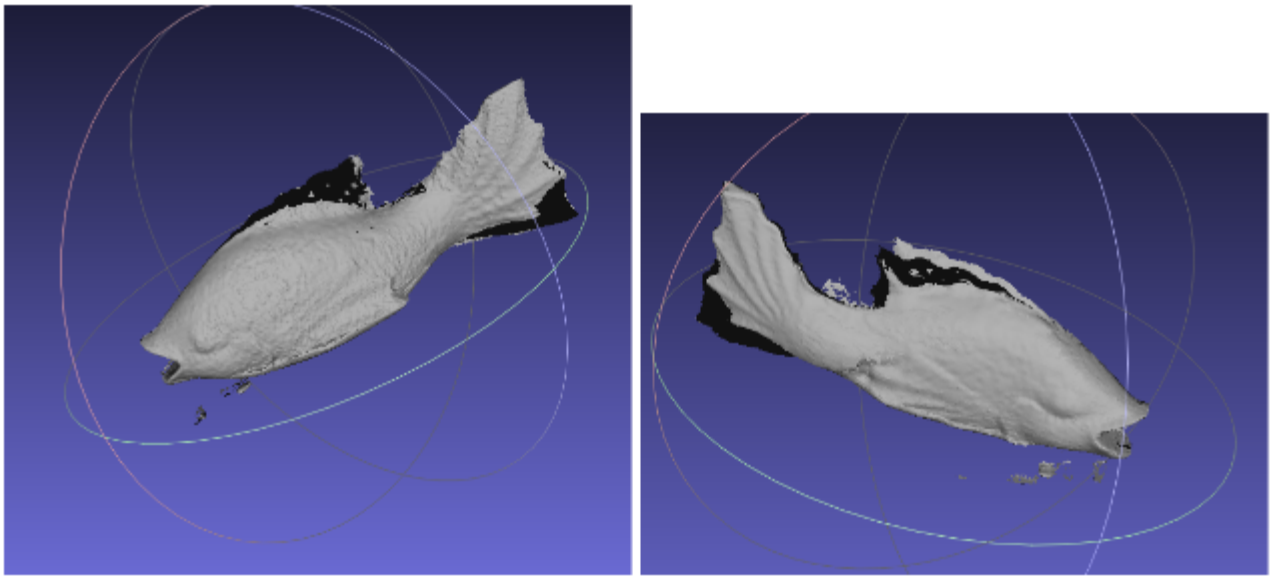
Results after Laplacian smoothing/ bounding-box pruning:



Final 3D Model:



Aligned 3D Model after Background Noise and Shadow Removal



Final 3D Model after Poisson Reconstruction

LIMITATION & IMPROVEMENTS

1. **Manual threshold adjustment**

- Required frequent trial-and-error due to varying lighting/noise conditions
- Depended on user intuition → reduced reproducibility

Planned Improvement:

- Implement adaptive thresholding based on brightness distribution
- Establish more controlled and consistent scan environments

2. **Manual point-click feature alignment**

- Subject to 1–2 pixel human error and slower processing
- User-dependent accuracy and scalability limitations

Planned Improvement:

- Automatic feature detection and alignment with verification steps

3. **Post-processing noise and shape smoothing issues**

- Residual noise occasionally remained after cleaning
- Over-smoothing damaged thin structures

Planned Improvement:

- Smarter filters preserving edges while removing noise
- Adaptive smoothing strength by region characteristics
- More automated and consistent post-processing

TAKEAWAYS

- 3Learned and implemented SVD-based multi-view alignment for point clouds
- Experienced mesh denoising and optimization techniques
- Gained strong understanding of Poisson surface reconstruction for watertight models
- Designed fully automated scanning pipeline with caching and batch stability
- Strengthened debugging and visualization competency in computer vision

- Enhanced understanding of spatial data and 3D geometry processing

Full experiment logs and detailed analysis are available in a separate report upon request.